

■ DevLabs × OpenLake IITBH

Session 1 — Git & GitHub

Web Development Bootcamp

■ Why Learn Web Development?



Freelancing

Earn money building websites for clients. Even small projects pay
■5K-50K+!



Open Source

Contribute to real projects used by millions. Build your reputation globally.



Build Your Startup

Turn your ideas into reality. No-code tools can't match custom dev.



Creativity

Build your own portfolio website and show the world what you can do!

■ **Fun Rule:** You have permission to update the group icon to random Web Dev memes! Keep it engaging!

■ Git vs GitHub — What's the Difference?

Git	GitHub
A command-line TOOL that runs on your computer	A WEBSITE to store and share your projects online
Like pip — runs commands, manages versions	Like Google Drive — but for your code!
Not a programming language. Not a framework.	Companies check your GitHub profile when hiring!

■ Explore GitHub — Key Concepts

■ Repository (Repo) Your project folder on GitHub. Contains all your code, history, and files.	■ Pull Request (PR) Ask the owner to merge your changes into their project. Core of open source!
■ Commit A snapshot of your code at a point in time. Like saving a game checkpoint!	■ Merge Combine changes from different branches into one. Teamwork in action!
■ Push Upload your local commits to GitHub. Sharing your work with the world!	■ Merge Conflict When two people edit the same line. Must be resolved manually!
■ Pull Download latest changes from GitHub to your computer.	■ Issues Bug reports, feature requests, or tasks. Public todo list for a project!
■ Fork Copy someone else's repository to your account. Start modifying their project!	■ README.md The first file people see. Written in Markdown. Your project's intro page!
■ Clone Download a GitHub repo to your local computer to work on it.	■ .gitignore Tell Git which files to ignore. Never push passwords or node_modules!

■■ Watch Merge Conflict explained: <https://www.youtube.com/shorts/79wdoaOrxRM>

■ Essential Git Commands

Command	What it does
git init	Initialize a new Git repository in current folder
git add .	Stage ALL changed files for commit
git add filename	Stage a specific file for commit
git commit -m 'message'	Save staged changes with a description
git status	See which files are changed/staged
git log	View commit history
git push origin main	Upload commits to GitHub main branch
git pull origin main	Download latest changes from GitHub
git clone	Download a GitHub repo to your computer

<code>git branch</code>	List all branches in repository
<code>git checkout -b feature</code>	Create and switch to a new branch
<code>git merge branch-name</code>	Merge another branch into current branch

■■ **Live Demo — Explore Bitefy!** A full-stack React + Django app built by a fellow IITian. See how React components, GitHub repos, commits, and branches look in a real project!

■ Your Tasks — Do These Today!

1 ■

Build a Great GitHub Profile

Add a profile picture, bio, location, and a pinned README. Show Jaswanth's profile as reference. Profile views counter is a nice touch!

2 ■

Push Your Existing Projects

Create repos for projects you've already built. Even small Python scripts count! `git init` → `git add .` → `git commit` → `git push`

3 ■

Play with Git Commands

Edit files in your repo, make commits, create branches. Try a pull request on someone else's repo — that's open source in action!